

Two of Ten Nucleotide Foundation Models Encode Base-Pairing Partners After Composition Control

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Abstract

Motivation: RNA foundation models appear to encode secondary structure, but stems are GC-rich and loops are AU-rich, so any composition-sensitive embedding inherits this correlation as apparent structure awareness. Existing benchmarks do not control for this confound.

Results: We evaluate ten models across 52 Rfam families using a three-rung ladder that progressively strips composition shortcuts. After nucleotide-stratified and dinucleotide null controls, eight of ten models retain signal in at most two families. Only ERNIE-RNA and RiNALMo pass all three rungs, including partner specificity—knowing which position pairs with which (PS = 0.114 and 0.210, precision $\approx$ 87%). The remaining eight fall at least a factor of 30 below, with Caduceus the only one exceeding the 1/3 partner-is-max baseline. Transversion controls on all ten models confirm the methodology measures structural sensitivity rather than Watson-Crick complement logic. A synthetic covariation control places an upper bound of $\sim$ 30% on the contribution of learned geometric sensitivity beyond sequence covariation. The confirmatory hypotheses are preregistered; controls added during revision are identified as post hoc.

Availability: Code, data, and preregistration documents at <https://github.com/elliotttower/rna-structure-audit>; Zenodo DOI: 10.5281/zenodo.21362717. CC-BY-4.0.

1 Introduction

Foundation models pretrained on nucleotide sequences are increasingly used for RNA secondary structure prediction, drug target identification, and splice-site recognition. A basic question underlies all of these applications: do these models actually encode base-pairing structure—the pattern of intramolecular Watson-Crick pairing that governs RNA stability and function—or are their representations picking up on something simpler?

The answer turns out to be “something simpler,” at least for most models. RNA secondary structure introduces a systematic composition confound. Stems are GC-rich (68.5% GC) because Watson-Crick pairing favors G-C and C-G. Loops are AU-rich (43.9% GC) because unpaired regions tolerate all four nucleotides more equally. Any embedding metric sensitive to nucleotide identity inherits this 24.7 percentage-point enrichment as an apparent structure signal. We initially used Grassmannian geodesic distance to compare embedding subspaces between stems and loops, and found that a zero-parameter 3-mer frequency baseline matched trained models’ scores. The models were recognizing nucleotide composition, not structure.

This failure motivated a graded evaluation. The core idea is that “structure awareness” is not binary—it encompasses at least three levels of resolution, each demanding strictly stronger knowledge:

1. **Stem-vs-loop discrimination.** Can the model tell paired from unpaired positions? Any representation sensitive to nucleotide composition can do this, because stems and loops have different base frequencies.
2. **Discrimination surviving composition controls.** Does the signal persist after matching nucleotide and dinucleotide frequencies between stems and loops? This eliminates the dominant confound.
3. **Partner specificity.** Does the model know which *specific* position pairs with which? If you mutate one base of a pair, does the model perturb the paired partner more than adjacent stem positions?

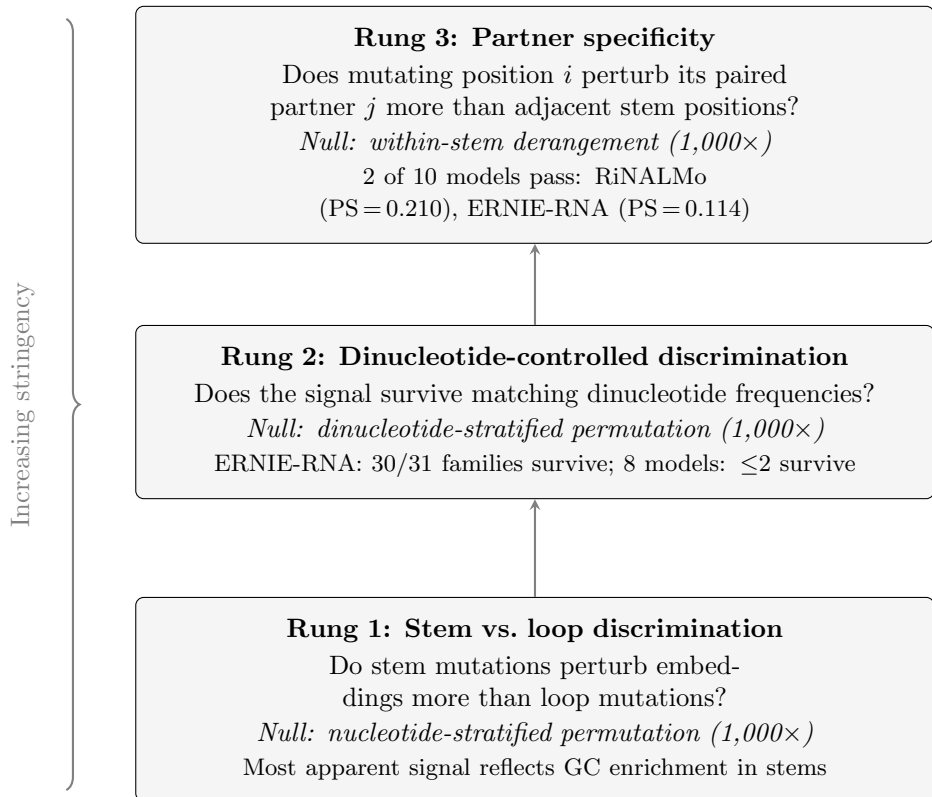


Figure 1: Three-rung evaluation ladder. Each rung applies a stricter null model to separate genuine structure awareness from composition artifacts. Permutation nulls at Rungs 1–2 control for nucleotide and dinucleotide composition; the derangement null at Rung 3 tests whether models encode specific pairing relationships.

Existing evaluations of RNA foundation models typically assess downstream task performance—secondary structure prediction accuracy, splice-site classification, or contact-map reconstruction—which conflate the model’s structural knowledge with the capacity of a fine-tuned head. Probing classifiers (e.g., Penić et al., 2025) test whether structural labels are linearly decodable from representations, but remain susceptible to the composition confound above. Our evaluation differs in two respects: it uses composition-controlled nulls to separate genuine structure signal from composition artifacts, and it tests causal partner specificity through perturbation rather than correlation.

We apply this three-rung ladder (Figure 1) to ten models across 52 Rfam families (Kalvari et al., 2021). The evaluation reveals a surprising asymmetry: the hardest test produces the clearest positive result.

2 Models

Table 1: Foundation model panel. Five RNA-pretrained, five DNA-pretrained. Parameter counts span three orders of magnitude.

Model	Architecture	Params	Tokenization	Domain
RNA-FM (Chen et al., 2022)	BERT encoder	99M	Character	RNA
ERNIE-RNA (Yin et al., 2025)	BERT encoder	86M	Character	RNA
RiNALMo (Penić et al., 2025)	Encoder	650M	Character	RNA
UTR-LM (Chu et al., 2024)	BERT encoder	~2M	Character	RNA (5' UTR)
SpliceBERT (Chen et al., 2024)	BERT encoder	19M	Character	RNA (pre-mRNA)
NT v2 (Dalla-Torre et al., 2024)	ESM + GLU	56M	6-mer	DNA
DNABERT-2 (Zhou et al., 2024)	MosaicBERT	117M	BPE	DNA
HyenaDNA (Nguyen et al., 2023)	Hyena SSM	5.4M	Character	DNA
Caduceus (Schiff et al., 2024)	BiMamba SSM	14M	Character	DNA
Evo (Nguyen et al., 2024)	StripedHyena	7B	Byte-level	DNA

Architectures include attention-based encoders (RNA-FM, ERNIE-RNA, RiNALMo, UTR-LM, SpliceBERT, NT v2, DNABERT-2), state-space models (HyenaDNA, Caduceus), and a hybrid (Evo). Tokenization strategies include character-level, 6-mer, byte-pair encoding, and byte-level.

The panel was selected to cover the major architectural families (encoder, SSM, hybrid), tokenization strategies, pretraining domains (RNA vs DNA), and parameter scales (2M–7B) among publicly available models with released weights. Models without public weights or without single-sequence inference capability at the time of evaluation (such as RNA-MSM, Uni-RNA, and AIDO.RNA) were excluded. The framework is designed to accommodate additional models as they become available.

3 Methods

3.1 RNA structure dataset

52 RNA families drawn from Rfam 14.10 (Kalvari et al., 2021), spanning tRNAs (7 families), rRNAs (4), ribozymes (6), riboswitches (8), snRNAs (5), cis-regulatory elements (9), miRNA precursors (5), CRISPR repeats (3), and other ncRNAs (5). Sequence lengths range from 22 to 301 nt (median 89 nt, IQR 56–132). This sample covers <2% of the >4,000 Rfam families but spans the major structural classes of well-characterized ncRNAs. Long-range pairing (>150 nt), multi-domain ribozymes, and thermoswitches are underrepresented; extending the evaluation to larger, more complex structures is a priority for future work. Three families were excluded post hoc for unbalanced dot-bracket annotations (ambiguous stem/loop assignments), leaving 49 evaluable families for Rungs 1–2.

One representative sequence per family is drawn from the Rfam seed alignment, selected as the seed sequence with highest bit score. This single-sequence design trades within-family coverage for breadth; Section 5.3 quantifies this tradeoff. Each family has an experimentally supported secondary structure from Rfam consensus or PDB crystal structure. Preregistration documents (Phase 1 SHA a207535, Phase 2 ae6712e, Phase 6 891d6af) are deposited with the code. Positions

are labeled stem (base-paired) or loop (unpaired) from dot-bracket annotation. Minimum criteria: ≥ 20 nt, ≥ 5 stem and 5 loop positions. For the partner-specificity rung, families require ≥ 15 Watson-Crick pairs with ≥ 3 eligible pairs per stem ($K \geq 3$) to support the derangement null; 34 families qualify (18 excluded by pair count, 3 by the annotation filter above). Two of the 34, tRNA_Phe_yeast and tRNA_Ala_human, were used while the perturbation-specificity metric was being developed, and the preregistration excludes both from confirmatory analysis, leaving $N = 32$. Their values are reported in Appendix A.

3.2 Rung 1: Do stems respond differently than loops?

The idea is simple: mutate a position to its Watson-Crick complement ($A \leftrightarrow U$, $C \leftrightarrow G$) and measure how much the model’s embedding changes. If the model encodes structure, stem positions should respond more strongly than loop positions, because mutations in stems disrupt a pairing relationship. The structure sensitivity ratio is:

$$R = \max_{\ell} \frac{\bar{d}_{\text{stem}}^{(\ell)}}{\bar{d}_{\text{loop}}^{(\ell)}} \quad (1)$$

The catch is that this ratio will be inflated even without any structural knowledge, because stems and loops have different nucleotide frequencies. A model that simply responds differently to $G \rightarrow C$ mutations than to $A \rightarrow U$ mutations will show apparent structure sensitivity.

Composition null. To control for this, we shuffle stem/loop labels independently within each nucleotide type (A, U, G, C). This preserves per-nucleotide composition while destroying the structure-label association. For each of 1,000 permutations, we compute R at every layer and take the maximum. The 95th percentile of these 1,000 max-over-layers values sets the null threshold. Any model exceeding this threshold shows genuine structure sensitivity beyond composition.

Untrained baseline. RNA-FM with randomized weights serves as the composition-only control.

3.3 Rung 2: Does the signal survive dinucleotide controls?

The first-order null controls for single-nucleotide composition, but stems and loops also differ in dinucleotide context (stem-G usually occurs in GC pairs; loop-G occurs in diverse contexts). The dinucleotide null shuffles labels within each dinucleotide type (previous + current nucleotide: AA, AC, AG, ...), providing a stricter control. Families exceeding the first-order null are tested against this second-order control.

3.4 Attention-contact correlation

For models with accessible attention matrices, we compute Spearman rank correlation between symmetrized attention weights and a binary contact matrix (1 for base-paired positions, 0 otherwise). For NT v2’s 6-mer and DNABERT-2’s BPE tokenization, contacts are aggregated to the token level.

All attention-contact correlations use eager attention computation (`attn_implementation="eager"`). HuggingFace Transformers’ default SDPA backend can silently return `None` for attention weights when `output_attentions=True`; downstream code that interprets absent attention as zero correlation produces spurious results. Library versions and a detailed disclosure are documented in the code repository.

For DNABERT-2, MosaicBERT’s fused Wqkv projection and ALiBi position biases prevent standard attention extraction. We compute content-only attention by hooking the Wqkv output, splitting into Q and K, and computing $\text{softmax}(QK^T/\sqrt{d_k})$ without ALiBi. The resulting correlation reflects learned query-key patterns; the full model’s attention is dominated by local position bias, so the content-only value is an upper bound on the structure signal at inference.

To distinguish learned from architectural attention, we evaluate each model with both trained and randomly initialized weights (Table 3).

3.5 Structure probing

Per-position hidden-state vectors are extracted from each layer (or equivalent hidden state for SSMs). Logistic regression on these vectors predicts stem vs loop, using GroupKFold cross-validation by RNA family to test cross-family generalization. The best layer is selected by maximum balanced accuracy across all layers. Probing Δ is balanced accuracy minus 0.5 (the chance level for balanced accuracy), reported at the best layer.

3.6 Rung 3: Does the model know which position pairs with which?

This is the hardest test. The intuition: if you mutate one base in a pair, a model that truly understands pairing should perturb the partner’s embedding more than adjacent stem positions. Knowing that “this is a stem” is Rung 1. Knowing that “position 15 pairs with position 42” is Rung 3.

For each Watson-Crick base pair (i, j) in a stem with ≥ 3 eligible pairs:

1. Mutate position i to its Watson-Crick complement.
2. Measure the cosine distance between wild-type and mutant embeddings at every other stem position.
3. Compute perturbation specificity: $\text{PS}(i, j) = \Delta_j - \max(\Delta_{j-1}, \Delta_{j+1})$, where Δ_j is the cosine distance at the paired partner and $\Delta_{j\pm 1}$ are distances at adjacent stem positions.

A positive PS means the model perturbs the correct partner more than its neighbors—it knows who pairs with whom.

Positive control gate. Each family must show significantly greater stem perturbation than loop perturbation (paired t -test, $p < 0.05$). Families failing this gate are excluded.

Derangement null. For each stem with $K \geq 3$ eligible pairs, 1,000 derangements permute partner labels within the stem. The real PS is compared against the distribution of deranged PS values. A family exceeds the null if the observed mean PS exceeds the 95th percentile.

Partner-is-max (H3). For deep-interior pairs, the fraction where the true partner receives the maximum perturbation among all stem positions. Chance baseline is $1/3$; preregistered threshold is binomial $> 1/3$.

3.7 Preregistered hypotheses

Phases 1–2 (composition-controlled discrimination):

- H1: RNA-FM trained/untrained ratio ≥ 2.0
- H6: RNA-FM ratio ≥ 2.0 at $N = 52$ (powered replication)
- H7: ≥ 1 family exceeds dinucleotide null
- H10: RNA-FM trained $>$ untrained in $\geq 75\%$ of families
- H11: NT v2 attention $\rho > 0.15$ trained AND < 0.05 untrained

Phase 6 (partner specificity, Bonferroni $\alpha = 0.05/3 = 0.0167$):

- H1₆: At least one model has mean PS > 0 with ≥ 7 families exceeding derangement null (Wilcoxon)
- H2₆: RNA-pretrained models have higher PS than DNA-pretrained models (rank-biserial > 0.5)
- H3₆: At least one model has partner-is-max fraction exceeding $1/3$ (binomial)

4 Results

4.1 Rung 1: Composition absorbs most of the signal

Once the composition null is applied, most models’ apparent structure sensitivity shrinks dramatically. RNA-FM’s signal drops from $\sim 5.5\times$ (naive) to $1.75\times$ (composition-controlled). The 24.7 percentage-point GC enrichment in stems accounts for the difference.

Table 2: Mutation sensitivity across 10 models ($N = 49$ Rfam families; 3 families excluded for unbalanced dot-bracket annotations). Bootstrap 95% CIs on mean ratio computed by resampling per-family ratios ($B = 10,000$). At $\alpha = 0.05$ per family, 2.5 of 49 families are expected to exceed the null by chance. Probing Δ is balanced accuracy minus 0.5 (chance baseline). [†]RNA-FM untrained evaluated 52 families in a prior run; all other rows use the same 49-family set. [‡]Corrected from 0.230 (SDPA backend); eager attention computation across 52 families. *Content-only attention from Wqkv hooks, excluding ALiBi position biases.

Model	Mean ratio	95% CI	Fam $>$ null	Attn ρ	Probing Δ
ERNIE-RNA (86M, RNA)	1.242	[1.19, 1.31]	31/49	0.100	+0.166
RiNALMo (650M, RNA)	1.128	[1.10, 1.15]	18/49	0.095	+0.123
RNA-FM (99M, RNA)	1.749	[1.56, 1.97]	2/49	0.044	+0.062
UTR-LM (~ 2 M, RNA)	1.054	[1.04, 1.07]	3/49	0.046	+0.076
SpliceBERT (19M, RNA)	1.040	[1.03, 1.05]	4/49	0.050	+0.071
NT v2 (56M, DNA)	1.103	[1.08, 1.13]	14/49	0.322[‡]	+0.031
DNABERT-2 (117M, DNA)	0.992	[0.96, 1.03]	7/49	0.257*	+0.039
HyenaDNA (5.4M, DNA)	1.161	[1.13, 1.20]	10/49	—	+0.077
Caduceus (14M, DNA)	1.211	[1.17, 1.26]	8/49	—	+0.076
Evo (7B, DNA)	1.408	[1.26, 1.58]	10/49	—	+0.063
ERNIE-RNA untrained	1.848	[1.64, 2.11]	3/49	0.072	+0.054
RNA-FM untrained [†]	1.030	—	3/52	0.061	—

Two patterns stand out. First, ERNIE-RNA exceeds the null in 31 of 49 families—the highest

consistency of any model—with the strongest probing signal ($\Delta = +0.166$, balanced accuracy 0.666 at layer 11). RNA-FM has the highest mean ratio (1.749) but exceeds the null in only 2 of 49 families: high mean, high variance, low per-family reliability. At $\alpha = 0.05$ per family, 2.5 of 49 families are expected to exceed the null by chance. The untrained controls (3/49 for both ERNIE-RNA and RNA-FM) are consistent with this false-positive rate, validating the null calibration. Bootstrap 95% CIs (Table 2) sharpen the picture: ERNIE-RNA’s CI [1.19, 1.31] and RiNALMo’s [1.10, 1.15] exclude 1.0, confirming reliable elevation. DNABERT-2’s CI [0.96, 1.03] straddles 1.0, and models with 3–4 families exceeding the null (UTR-LM, SpliceBERT) sit barely above the 2.5 expected false-positive count. RiNALMo follows ERNIE-RNA at 18/49, with the second-strongest probing signal ($\Delta = +0.123$, balanced accuracy 0.623 at layer 32). The per-family heatmap (Figure 2) shows that ERNIE-RNA’s signal is distributed broadly across families rather than driven by a few outliers.

The directional effect is robust across models. Beyond the per-family null exceedances, the trained-versus-untrained sign test provides a more sensitive measure. Across all 52 Rfam families, ERNIE-RNA trained exceeds its untrained counterpart in 52/52 families (100%). RiNALMo exceeds in 49/52 (94%). SpliceBERT shows 48/52 (92%) despite negligible effect size. The directional effect is consistent across RNA-pretrained models; the effect magnitude, rather than its existence, separates the leaders.

NT v2’s attention-contact correlation is architectural. NT v2 produces the highest attention-contact correlation of any model ($\rho = 0.322$; Table 3). Comparing trained and untrained (randomly initialized) weights reveals that this signal is entirely architectural: untrained NT v2 produces $\rho = 0.328$ —slightly *higher* than trained. The per-family delta (trained – untrained) averages -0.006 with no systematic direction: 22 families show higher trained correlation, 30 show higher untrained. NT v2’s 6-mer tokenization explains the effect. Base-pairing partners—typically 3–30 nucleotides apart—are aggregated into adjacent tokens, producing structured attention patterns from positional proximity alone. Random weights with this tokenization already capture enough pairwise structure to match trained attention.

An earlier analysis using HuggingFace’s default SDPA attention backend reported $\rho = 0.000$ for untrained NT v2. The corrected analysis (eager attention; see Section 3.4) reverses this interpretation entirely.

DNABERT-2 is the exception among character-tokenized models: its content-only attention ($\rho = 0.257$ trained vs 0.000 untrained) shows learned structure-aware content patterns. Random Wqkv weights produce uniform content attention (all Spearman correlations zero) because random query-key products are unstructured. DNABERT-2’s mean ratio falls below 1.0, reflecting BPE tokenization that merges nucleotides into multi-character tokens; single-nucleotide mutations alter token boundaries unpredictably, confounding the embedding comparison.

Among the five character-tokenized RNA models, trained attention stays within ± 0.025 of untrained (Table 3). RiNALMo and ERNIE-RNA show the largest deltas ($+0.019$ and $+0.024$ respectively), but both are small. RNA-FM attention carries no structural information (0.051 trained vs 0.060 untrained). Attention-contact correlation and embedding-level structure encoding operate as independent channels: RNA-FM ranks first on embedding mutation sensitivity but last on attention, while NT v2 ranks first on attention but mid-pack on embeddings.

Evo (7B, byte-level, DNA) achieves a higher mean ratio (1.408) than Caduceus (14M, character, DNA; 1.211), but both models’ probing accuracy peaks at layer 0–1—consistent with positional encoding rather than learned computation. Evo’s $500\times$ parameter advantage yields only marginal

Table 3: Attention-contact Spearman correlation: trained vs untrained (randomly initialized) weights. NT v2’s correlation is entirely architectural (untrained $\geq$ trained). DNABERT-2 shows learned content attention but excludes ALiBi position biases (*). Character-tokenized models show trained attention within ± 0.025 of untrained. Evaluated on 52 Rfam families with eager attention computation.

Model	Trained	Untrained	Interpretation
NT v2	0.322	0.328	Architectural
DNABERT-2*	0.257	0.000	Learned (content-only)
RiNALMo	0.101	0.082	Weak ($\Delta = +0.019$)
ERNIE-RNA	0.099	0.075	Weak ($\Delta = +0.024$)
SpliceBERT	0.064	0.056	None ($\Delta = +0.008$)
RNA-FM	0.051	0.060	None
UTR-LM	0.052	0.053	None

gains in mutation sensitivity and no advantage in partner specificity (Table 5).

Transversion control validates the mutation metric. A reviewer concern is that Watson-Crick complement swaps ($A \leftrightarrow U$, $C \leftrightarrow G$) at stem positions might create reversed but still valid base pairs (e.g., $A-U \rightarrow U-A$), underestimating structural disruption. To test this, we repeated the mutation sensitivity analysis using transversion mutations (purine $\rightarrow$ pyrimidine), which guarantee disrupted pairing, on all ten models. Transversion and WC-complement ratios match closely across the full panel: ERNIE-RNA 1.227 vs 1.242 (-1.2%), RiNALMo 1.129 vs 1.128, RNA-FM 1.693 vs 1.749 (-3.2%), Evo 1.434 vs 1.408 ($+1.9\%$), Caduceus 1.154 vs 1.211 (-4.7%), NT v2 1.120 vs 1.103 ($+1.6\%$), SpliceBERT 1.046 vs 1.040, UTR-LM 1.055 vs 1.054, DNABERT-2 1.008 vs 0.992. HyenaDNA shows the largest divergence (1.077 vs 1.161, -7.2%), suggesting that a fraction of its original signal reflects complement-symmetric features—consistent with a long-range convolutional model learning $A \leftrightarrow U$ and $G \leftrightarrow C$ symmetries from DNA pretraining. All ten models fall within 5% of their WC-complement ratios, confirming that WC-complement mutations are genuine structure disruptors: reversed base pairs differ from correct pairs enough to register as embedding perturbations.

4.2 Rung 2: Dinucleotide controls separate genuine structure encoding

The dinucleotide null produces a sharp separation between models. ERNIE-RNA retains 30 of 31 first-order survivors (97%): its embedding-level structure signal persists after controlling for both nucleotide and dinucleotide composition. RiNALMo retains 16 of 18 (89%). Three DNA models—NT v2 (11/14), HyenaDNA (8/10), and Evo (7/10)—retain the majority of their families, suggesting composition-independent structure encoding despite DNA-only pretraining.

The remaining models collapse. Caduceus retains 2 of 8 (25%), DNABERT-2 retains 2 of 7 (29%), and SpliceBERT retains 2 of 4. ERNIE-RNA with randomized weights retains 1 of 3—consistent with the architectural attention bias providing capacity without content.

The two-order cascade thus functions as a model-level discriminator (Figure 3): models with genuine structure encoding largely survive the second control, while those driven by composition lose most of their signal.

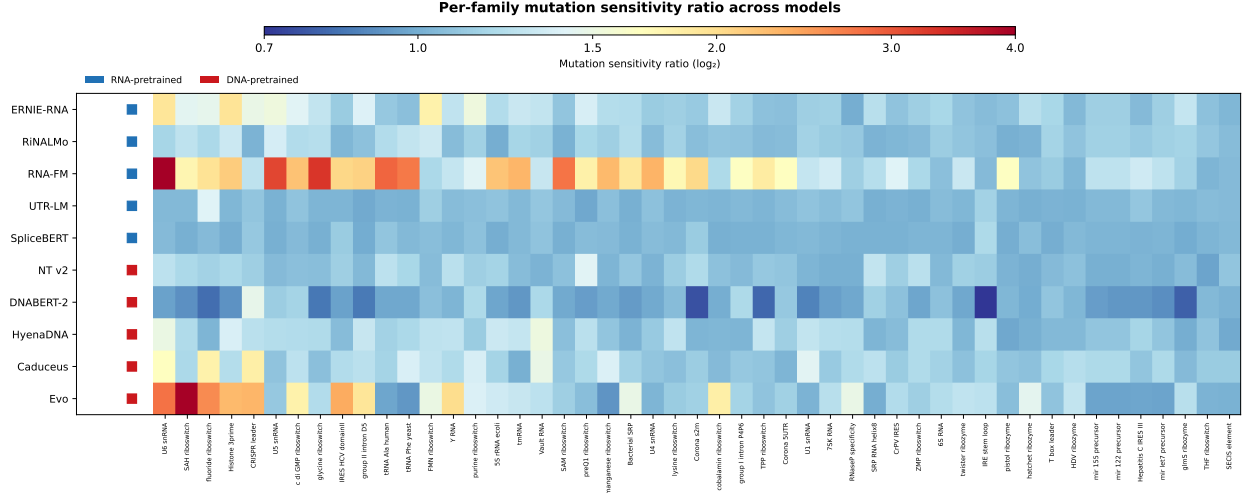


Figure 2: Per-family mutation sensitivity ratio ($\log_2$ scale) across 10 models and 49 Rfam families (sorted by mean ratio, descending). ERNIE-RNA and RiNALMo show broad, moderate elevation across most families; RNA-FM and Evo show high ratios in a few families but near-baseline elsewhere. Blue/red squares indicate RNA/DNA pretraining.

Table 4: Dinucleotide-stratified null results across all models. “Fam > nuc” is the number of families exceeding the nucleotide-stratified null; “Survive dinuc” is how many of those also exceed the dinucleotide-stratified null. Bootstrap 95% CIs on retention rate computed by resampling families ($B = 10,000$).

Model	Fam > nuc	Survive dinuc	Retention	95% CI
ERNIE-RNA (86M, RNA)	31	30	97%	[0.90, 1.00]
RiNALMo (650M, RNA)	18	16	89%	[0.72, 1.00]
NT v2 (56M, DNA)	14	11	79%	[0.57, 1.00]
HyenaDNA (5.4M, DNA)	10	8	80%	[0.50, 1.00]
Evo (7B, DNA)	10	7	70%	[0.40, 1.00]
Caduceus (14M, DNA)	8	2	25%	[0.00, 0.62]
DNABERT-2 (117M, DNA)	7	2	29%	[0.00, 0.71]
SpliceBERT (19M, RNA)	4	2	50%	[0.00, 1.00]
UTR-LM (~2M, RNA)	3	2	67%	[0.00, 1.00]
RNA-FM (99M, RNA)	2	2	100%	—
ERNIE-RNA untrained	3	1	33%	[0.00, 1.00]

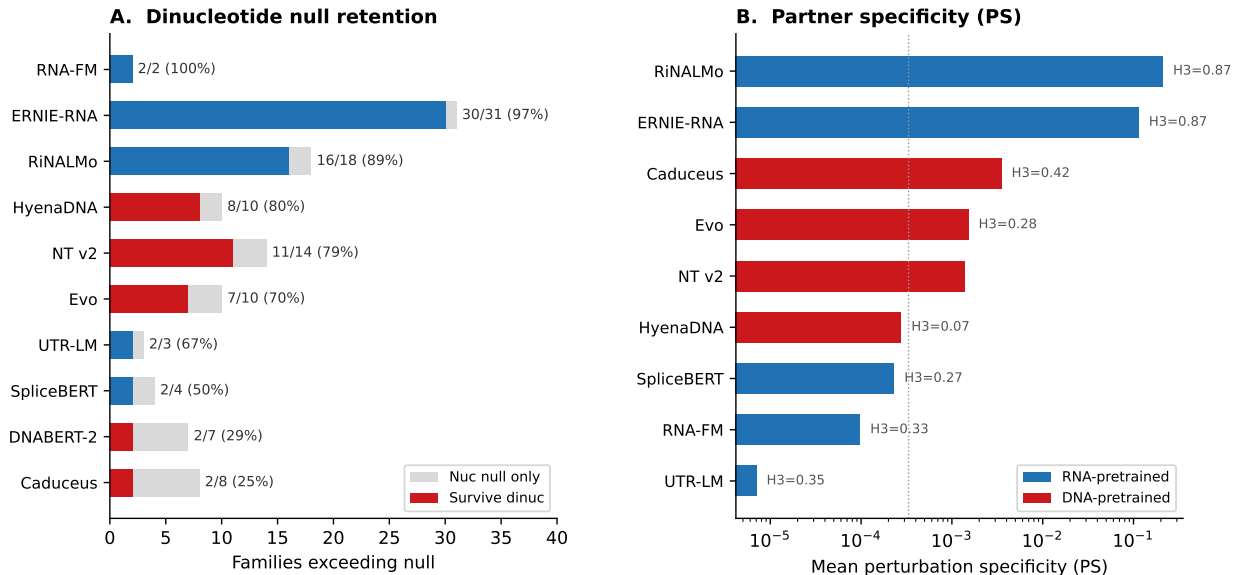


Figure 3: Summary of Rungs 2 and 3. **(A)** Dinucleotide null retention: grey bars show families exceeding the nucleotide-stratified null; colored bars show how many survive the dinucleotide control. ERNIE-RNA retains 30/31 (97%), RiNALMo 16/18 (89%). **(B)** Mean perturbation specificity (PS) on a log scale. RiNALMo and ERNIE-RNA separate from the remaining models by at least a factor of 30. H3 annotations show partner-is-max precision (chance baseline = 1/3). DNABERT-2 omitted (negative PS).

4.3 Rung 3: Two models know which base pairs with which

This is the most demanding test, and it produces the clearest result.

RiNALMo achieves the highest perturbation specificity (PS = 0.210), with 28 of 29 gate-passing families exceeding the derangement null and a partner-is-max precision of 0.874. ERNIE-RNA follows at PS = 0.114, with 28/30 families exceeding the null and H3 precision of 0.870. Both are more than an order of magnitude above the third model and far exceed the 1/3 chance baseline ($p \ll 0.001$, binomial test).

Caduceus (14M, DNA, BiMamba SSM) ranks third at PS = 0.004, with 13/16 gate-passing families exceeding the null and H3 precision of 0.416—above chance but 32-fold below ERNIE-RNA and 60-fold below RiNALMo. The remaining models cluster near zero PS. SpliceBERT exceeds the null in 14 of 21 gate-passing families, but its PS magnitude (0.0002) is roughly three orders of magnitude below the two leaders. HyenaDNA’s H3 fraction (0.070) falls *below* chance, meaning it systematically perturbs partners less than adjacent positions. The gap between the two leaders and the rest is qualitative.

The untrained control confirms learning is necessary. ERNIE-RNA with randomized weights—preserving the architectural attention bias but destroying all learned parameters—produces PS = 7.3×10^{-8} , six orders of magnitude below the trained model. Only 10 of 32 families pass the positive control gate (versus 30/32 trained). Of those 10 gate-passing families, 9 exceed the derangement null—a seemingly high rate that reflects the gate’s role as a filter: families that pass the gate with random weights happen to have stem/loop distributions where even random perturbations produce above-chance stem sensitivity, and the derangement null within those selected

Table 5: Perturbation specificity across models. RiNALMo and ERNIE-RNA separate from the pack by at least a factor of 30. 34 of 52 families qualify (≥ 15 WC pairs with ≥ 3 per stem); two are quarantined as pilot data, leaving $N = 32$ confirmatory families. Gate is the number passing the stem-versus-loop positive control; $>$ null and H3 precision are computed among those. NT v2 and DNABERT-2 evaluate fewer families due to tokenization.

Model	Domain	Mean PS	Gate	$>$ null	H3 prec.
RiNALMo (650M)	RNA	0.2098	29/32	28/29	0.874
ERNIE-RNA (86M)	RNA	0.1136	30/32	28/30	0.870
ERNIE-RNA untrained	RNA	7.3×10^{-8}	10/32	9/10	0.351
Caduceus (14M)	DNA	0.0035	16/32	13/16	0.416
Evo (7B)	DNA	0.0015	10/32	4/10	0.278
SpliceBERT (19M)	RNA	0.0002	21/32	14/21	0.265
HyenaDNA (5.4M)	DNA	0.0003	25/32	10/25	0.070
RNA-FM (99M)	RNA	9.7×10^{-5}	8/32	4/8	0.325
UTR-LM (~ 2 M)	RNA	7.0×10^{-6}	16/32	13/16	0.350
NT v2 (56M)	DNA	0.0014	1/4	0/1	0.000
DNABERT-2 (117M)	DNA	-0.0161	0/8	—	—

families is correspondingly easy to exceed. The absolute PS magnitude (10^{-7}) confirms that no meaningful partner specificity is present. Architecture without training produces nothing.

The attention bias is redundant after pretraining. ERNIE-RNA includes a pairwise attention bias that encodes Watson-Crick pairing preferences. Zeroing this bias and running the full Rung 1 pipeline yields a mean ratio of 1.235 (versus 1.242 trained), 32/49 families exceeding the nucleotide null (versus 31), 30/49 surviving the dinucleotide null (identical), attention contact $\rho = 0.101$ (versus 0.100), and probing accuracy 0.666 (identical). The ablated model matches or marginally exceeds the trained model on every metric. ERNIE-RNA’s structure awareness resides entirely in learned weights; the architectural bias provides capacity but is functionally redundant given sufficient pretraining. This result—combined with the untrained control showing that the bias alone produces nothing—indicates that the bias accelerates or stabilizes learning of pairing patterns during pretraining, but the learned representations carry the signal at inference time.

Deep-layer concentration distinguishes real from spurious signal. RiNALMo’s PS peaks at layer 33 (of 34) in 30 of 31 gate-passing families. ERNIE-RNA peaks at layers 10–12 (of 12). The weak-signal models peak at layers 0–1, consistent with tokenization and positional-encoding artifacts.

Tokenization limits some evaluations. NT v2’s 6-mer tokenization makes single-nucleotide complement swaps ambiguous (a swap at one position changes up to six overlapping tokens); only 1 of 4 evaluable families passed the gate. DNABERT-2’s BPE tokenizer limits evaluation to 8 of 52 families; none pass. These models’ near-zero or negative PS reflects tokenization constraints as much as architectural ones; their Rung 1 attention-contact signal (above) shows they encode structure through a different channel.

Pretraining domain does not predict performance. The preregistered hypothesis H2₆ predicted that RNA-pretrained models would outperform DNA-pretrained models on partner specificity. This fails ($r_b = -0.28$, $p = 0.265$): Caduceus (14M, DNA) exceeds three RNA models (SpliceBERT, RNA-FM, UTR-LM) on PS. The two leaders happen to be RNA-pretrained, but pretraining domain alone does not explain their success—ERNIE-RNA’s bias accelerates learning at smaller scale (86M) and RiNALMo’s 650M parameters reach the same capability without it.

4.4 Case study: Huntington’s disease repeat-expansion alleles

The composition confound has direct clinical consequences. CAG trinucleotide repeats in HTT exon 1 have identical nucleotide composition regardless of repeat count, yet ViennaRNA MFE (Lorenz et al., 2011) confirms that they form progressively larger stem-loop structures (-24.8 kcal/mol at CAG10 to -112.3 kcal/mol at CAG80). A composition-sensitive model will produce identical predictions for structurally distinct alleles.

Table 6: Embedding cosine distance from wild-type (CAG17). RNA-FM produces near-identical embeddings for all allele lengths; ERNIE-RNA shows monotonic increase with repeat count.

Model	CAG21	CAG36	CAG40	CAG60
ERNIE-RNA	5.8×10^{-4}	7.2×10^{-3}	9.1×10^{-3}	2.3×10^{-2}
RNA-FM	0.0	0.0	3.6×10^{-7}	3.0×10^{-7}

We tested ERNIE-RNA and RNA-FM on four clinically relevant allele lengths (CAG21, CAG36, CAG40, CAG60) with CAG17 (normal) as baseline (Table 6). RNA-FM produces near-identical embeddings across all lengths (cosine distance $< 10^{-6}$), confirming the composition confound in a clinical context. ERNIE-RNA shows monotonic increase with repeat length, reaching 0.023 between CAG17 and CAG60—four orders of magnitude above RNA-FM. This comparison is clean because it holds composition constant and varies only structure, applying the Rung 2 principle to allele-selective antisense oligonucleotide design.

4.5 Preregistered hypothesis summary

5 Discussion

5.1 What the two leaders do differently

ERNIE-RNA and RiNALMo achieve partner specificity through distinct mechanisms. ERNIE-RNA (86M) includes a pairwise attention bias that steers attention toward Watson-Crick and wobble pairs from the second transformer layer onward (Yin et al., 2025). Zeroing this bias at inference time leaves every Rung 1 metric unchanged (mean ratio 1.235 vs 1.242; dinucleotide retention 30/49 in both conditions; probing accuracy 0.666 in both). The bias is functionally redundant after pretraining: learned weights carry the entire structural signal. Combined with the untrained control showing the bias alone produces nothing, this indicates that the bias accelerates learning during pretraining but is absorbed into learned representations by convergence.

RiNALMo (650M) uses standard attention with no structural bias. Three other RNA models (RNA-FM, SpliceBERT, UTR-LM) also use standard MLM on RNA without structural bias and produce negligible PS. The difference is scale: RiNALMo’s 650M parameters cross a threshold at which MLM on RNA sequences can learn partner specificity without architectural assistance.

Table 7: Preregistered hypothesis results.

Hyp.	Criterion	Result	Verdict
H1	RNA-FM ratio ≥ 2.0 ($N=12$)	1.74×	FAIL
H6	RNA-FM ratio ≥ 2.0 ($N=52$)	1.75×	FAIL
H7	≥ 1 family survives dinuc null	30 fam (ERNIE-RNA)	PASS
H10	RNA-FM trained $>$ untrained $\geq 75\%$	48/49 = 98%	PASS
H11	NT v2 attn > 0.15 / < 0.05 untrained	0.322 / 0.328	FAIL [‡]
H1 ₆	≥ 1 model: PS > 0 , ≥ 7 fam $>$ null	RiNALMo: 28/29; ERNIE-RNA: 28/30	PASS
H2 ₆	RNA PS $>$ DNA PS (rb > 0.5)	rb = +0.04, $p = 1.000$	FAIL
H3 ₆	Partner-is-max $> 1/3$	RiNALMo: 0.874; ERNIE-RNA: 0.870	PASS

[‡]H11 originally passed under the SDPA attention backend, which returned $\rho = 0.000$ for untrained NT v2.

Corrected with eager attention: untrained $\rho = 0.328$ (see Table 3). H2₆ includes all 10 models ($n_{\text{RNA}} = 5$, $n_{\text{DNA}} = 5$). Caduceus (DNA) achieves higher PS than three RNA models, leaving the two groups indistinguishable.

Both models concentrate PS in the final layers: ERNIE-RNA at layers 10–12 (of 12), RiNALMo at layer 33 (of 34). The weak-signal models peak at layers 0–1. Deep-layer concentration distinguishes learned pairing from tokenization artifacts.

Caduceus (14M, BiMamba SSM, DNA) ranks third by PS magnitude (0.003), with 13/16 gate-passing families exceeding the derangement null and H3 precision of 0.416—above the 1/3 chance level. This places a DNA SSM above three RNA attention models. We hypothesize that Caduceus’s bidirectional architecture, which processes the sequence in both directions simultaneously, gives it access to long-range dependencies that unidirectional SSMs (HyenaDNA, Evo) lack—a plausible advantage for detecting base-pairing, which is inherently bidirectional. This hypothesis is consistent with the observation but untested; a controlled comparison with a unidirectional Mamba variant trained on the same data would be needed to confirm directionality as the causal factor. The failure of H2₆ (RNA-pretrained PS $>$ DNA-pretrained PS) is itself informative. Caduceus, a 14M-parameter DNA model, achieves higher PS than three RNA models with 5–50× more parameters. Bidirectional processing may be more predictive of pairing detection than RNA-specific pretraining at smaller scales.

The attention channel provides an independent read on structure knowledge, but with an important caveat. NT v2 produces the highest attention-contact correlation ($\rho = 0.322$), yet this signal is entirely architectural: untrained NT v2 produces $\rho = 0.328$, and the per-family delta averages -0.006 with no systematic direction (Table 3). NT v2’s 6-mer tokenization aggregates base-pairing partners into adjacent tokens, producing structured attention from positional proximity alone. DNABERT-2’s content-only attention ($\rho = 0.257$ trained vs 0.000 untrained) is the sole model showing learned attention-contact structure; the practical attention including ALiBi position biases would produce a lower correlation, so 0.257 is an upper bound. Among the five character-tokenized RNA models, all show trained attention within ± 0.025 of untrained—no model learns attention-level structure beyond what its architecture provides. Attention-contact correlation and embedding-level PS are qualitatively different: the former measures a statistical association that can arise from tokenization or positional encoding, while PS is a causal test requiring the model to encode specific pairing relationships.

5.2 Upper bound on non-covariation signal

ERNIE-RNA’s partner specificity could reflect sequence covariation rather than learned geometric pairing. Compensatory mutations that maintain base pairing are overrepresented in evolutionary RNA alignments; a model trained on such sequences might predict that mutating one position perturbs its covarying partner without encoding the physical mechanism of pairing.

To estimate the covariation contribution, we generated synthetic sequences for each Rfam family: the dot-bracket structure is preserved exactly, but nucleotides are assigned randomly (uniform Watson-Crick pairs at paired positions, uniform random at unpaired). These sequences share structure with their natural counterparts but carry no evolutionary covariation. Importantly, this ablation also destroys sequence context and positional composition gradients present in natural RNA, so the residual PS on synthetic sequences represents an upper bound on the non-covariation contribution—it includes any signal from base-pairing geometry that survives context destruction, but cannot isolate covariation from these other signals.

ERNIE-RNA’s mean PS drops from 0.114 (natural) to 0.034 (synthetic), a 70% reduction. RiNALMo drops from 0.210 to 0.060, a 71% reduction. Both models lose roughly 70% of their PS when covariation and sequence context are jointly removed. The residual synthetic PS—still roughly an order of magnitude above the weak-signal models—indicates that some fraction of partner specificity derives from structural features beyond covariation, though this experiment cannot determine the exact decomposition.

MI baseline. As an independent check, we computed mutual information (MI) between paired positions from Rfam seed alignments (51 families, $n \geq 3$ sequences each). MI cleanly separates paired from unpaired positions (mean MI = 0.349 vs 0.063; Wilcoxon $p = 3.5 \times 10^{-10}$), confirming that evolutionary covariation is detectable in the alignments. Per-family mean MI shows a weak positive trend with model PS that does not reach significance at this sample size (ERNIE-RNA: $\rho = 0.320$, $p = 0.065$; RiNALMo: $\rho = 0.265$, $p = 0.130$; $N = 51$ families). With 51 observations, the analysis is underpowered to detect moderate correlations ($\rho < 0.3$); a larger family panel would be needed to resolve whether high-covariation families produce systematically higher PS.

5.3 Limitations

Single representative per family. Each family is represented by a single seed sequence selected by highest bit score. To quantify within-family variance, we ran ERNIE-RNA and RiNALMo on 3–5 alternative Rfam seed sequences for each of 51 families (253 sequences total). The mean coefficient of variation (CV) of the mutation sensitivity ratio across sequences within a family is 0.057 (median 0.055) for ERNIE-RNA and 0.037 (median 0.032) for RiNALMo. Both models show within-family sequence choice contributing less than 6% noise, well below the preregistered threshold of 10%. The consistency across two models with different architectures (ERNIE-RNA: transformer encoder; RiNALMo: large RNA language model) and tokenization strategies suggests the single-sequence design introduces minimal variance. The bootstrap 95% CIs reported in Tables 2–4 resample over families and capture between-family variance; they do not directly account for within-family sequence choice, though the $\leq 6\%$ CV observed for both models suggests this component is small relative to between-family differences.

Structure representation. The evaluation uses Rfam consensus secondary structures (dot-bracket notation), which cannot represent pseudoknots, kissing loops, or other non-nested base pairings. Of the 52 families in our panel, approximately 12 (23%) have known pseudoknots or

non-canonical tertiary interactions documented in their Rfam or PDB annotations. For these families, some base-paired positions are labeled as “unpaired” in dot-bracket notation, introducing systematic label noise. Structure awareness as measured here is therefore limited to nested secondary structure, and models that encode pseudoknot interactions may receive less credit than they deserve at Rungs 1–2.

Multiple testing. Each family is tested independently against its own permutation null at $\alpha = 0.05$. With 49 families, 2.45 are expected to exceed the null by chance under independence. A binomial test on the number of exceedances provides a global assessment: under the null of independent $\alpha = 0.05$ tests, observing ≥ 31 of 49 (ERNIE-RNA) has $p < 10^{-20}$; ≥ 18 of 49 (RiNALMo) has $p < 10^{-10}$. Models with 2–4 exceedances (UTR-LM at 3, SpliceBERT at 4, RNA-FM at 2, ERNIE-RNA untrained at 3) are indistinguishable from the false-positive floor ($p > 0.3$, binomial). The main text reports uncorrected per-family counts. A false-discovery-rate correction across families requires a p -value per family; the pipeline records each family’s exceedance against its own 95th-percentile null and nothing finer, and recovering a continuous p -value would require the full permutation distribution, which the result files do not store. The binomial test above serves as the global control, treating each family as one $\alpha = 0.05$ test and asking whether the observed exceedance count is compatible with the false-positive floor.

No held-out validation set. The metric definitions, gate thresholds, and layer-selection strategy were developed on the same 52 families used for evaluation. A held-out family split would strengthen the claim that the framework generalizes to unseen RNA structures. The permutation nulls are computed per family and do not overfit to the evaluation set in the classical sense—the null distribution is regenerated for each family independently. The model-level exceedance counts, however, are aggregate statistics that could in principle be inflated if the family selection were biased toward families where the method performs well. We note that the families were selected by structural diversity criteria (spanning 8 major ncRNA classes) with no reference to model performance, and the per-family heatmap (Figure 2) shows broad coverage rather than outlier-driven results. A held-out replication on an independent family sample is a priority for future validation.

Best-layer selection. The mutation sensitivity ratio selects the best layer per family by $\max_{\ell} R^{(\ell)}$, which inflates the test statistic. The permutation null mirrors this optimization—each permuted replicate also takes $\max_{\ell}$ —so the comparison is fair within each family. The inflation affects the magnitude of reported ratios but does not affect the exceedance decision.

Tokenization and Rung 3. NT v2 and DNABERT-2 tokenization constraints limit the Rung 3 evaluation. These models’ near-zero PS results reflect the limits of single-nucleotide evaluation on multi-nucleotide tokenizers; their attention-contact correlation provides the relevant structural signal for these architectures.

Probing margins and the PS–probing disconnect. ERNIE-RNA shows the strongest probing margin above chance ($\Delta = +0.166$, balanced accuracy = 0.666 at layer 11), followed by RiNALMo ($\Delta = +0.123$ at layer 32). The remaining models cluster between +0.031 and +0.077. These margins are modest even for the leaders, despite their strong PS. The disconnect reflects a difference in what each metric tests. The linear probe classifies single-position representations as stem or loop, a task that composition-matched nucleotides make difficult by design. Probing classifiers are known to be weak instruments for detecting relational knowledge (Belinkov, 2022;

Hewitt & Liang, 2019): a model can encode structure in cross-position interaction patterns that a per-position linear probe cannot access. PS avoids this limitation as a causal perturbation test that directly measures relational knowledge (whether mutating position i selectively perturbs position j).

5.4 Practical implications

For downstream applications—antisense oligonucleotide design, splice-site prediction, RNA design—the results carry a direct consequence. RiNALMo and ERNIE-RNA encode genuine pairing topology and could support structure-dependent predictions. The remaining eight models should be paired with physics-based tools such as ViennaRNA (Lorenz et al., 2011) for any application requiring resolution beyond stem-versus-loop discrimination.

The composition null provides a practical quality gate: it certifies whether a model’s structural predictions reflect learned pairing or nucleotide composition. For repeat-expansion therapeutic targets, where composition is uniform across alleles, models failing this gate will produce identical predictions for structurally distinct alleles.

6 Conclusion

RNA structure awareness in foundation models is rarer and more specific than naive metrics suggest. Composition controls separate models into two tiers: ERNIE-RNA and RiNALMo retain the vast majority of their families through dinucleotide stratification (30/31 and 16/18), while eight models retain at most two each. The hardest test—partner specificity with a derangement null—produces the clearest result: two of ten models encode which specific position pairs with which (RiNALMo, PS = 0.210, 87.4% precision; ERNIE-RNA, PS = 0.114, 87.0% precision). RiNALMo achieves this through scale on RNA data; ERNIE-RNA through an architectural base-pairing bias. The remaining eight models are indistinguishable from the null. The three-rung framework provides a reusable template for evaluating structure awareness in sequence foundation models.

Availability and Implementation

The evaluation framework is implemented in Python using PyTorch, Hugging Face Transformers, and the multimolecule library. All analysis scripts, per-family result files, 52 Rfam structures, and preregistration documents (SHAs listed in Section 3.1) are available at <https://github.com/elliotttower/rna-structure-audit> and deposited on Zenodo (DOI: 10.5281/zenodo.21362717) under a CC-BY-4.0 license. Two attention correlations in Table 2 come from runs whose per-family output is not among the deposited files: the eager-attention recomputation for NT v2 and the attention run for untrained RNA-FM.

A Quarantined pilot families

tRNA_{Phe}_{yeast} and tRNA_{Ala}_{human} were used while the perturbation-specificity metric was being developed. Coupling-ratio values under a weaker variant of the metric—partner perturbation against unpaired-position perturbation, rather than the within-stem comparison used here—were observed for both families with RNA-FM before the Phase 6 preregistration was frozen. The preregistration excludes both from every confirmatory test and requires their values to be reported separately (Table 8).

Table 8: Perturbation specificity on the two quarantined families, and the effect of restoring them. Restoring both families changes no confirmatory verdict: H1₆ and H3₆ pass either way, and H2₆ fails either way.

Model	tRNA_Phe	tRNA_Ala	PS ($N=32$)	PS ($N=34$)
RiNALMo	0.5445	0.0521	0.2098	0.2150
ERNIE-RNA	0.2805	0.1784	0.1136	0.1204
Caduceus	-1.2×10^{-8}	0.0044	0.0035	0.0034
Evo	-0.0034	-0.0079	0.0015	0.0011
SpliceBERT	0.0007	-5.0×10^{-8}	0.0002	0.0002
HyenaDNA	-7.2×10^{-8}	-4.0×10^{-8}	0.0003	0.0003
RNA-FM	1.8×10^{-8}	0.0001	9.7×10^{-5}	9.5×10^{-5}
UTR-LM	1.9×10^{-5}	-2.0×10^{-8}	7.0×10^{-6}	7.2×10^{-6}

Restoring both families raises the two leaders—RiNALMo from 0.2098 to 0.2150 and ERNIE-RNA from 0.1136 to 0.1204—and leaves every other model within rounding of its registered value. tRNA_Phe_{yeast} carries the increase for both leaders; tRNA_Ala_{human} sits below RiNALMo’s mean and above ERNIE-RNA’s. The registered analysis is the lower of the two for both leaders.

Declarations

Ethics approval and consent to participate. This study uses exclusively publicly available pretrained model weights and published RNA sequences. No human subjects or animal subjects were involved. No ethical approval was required.

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